

Cybersecurity Threat Analytics

AI-Powered Cybersecurity Analytics & Decision Support Platform

Project Overview

Cybersecurity Threat Analytics is a **Data Analytics-based web application** that helps organizations monitor, analyze, detect, predict, and understand cyber threats using **Data Analytics, Machine Learning, and Artificial Intelligence**.

Unlike traditional intrusion detection systems that only classify traffic as **Normal** or **Attack**, this platform provides **complete cybersecurity intelligence**.

It answers questions like:

- What attacks are occurring?
- Why did they happen?
- Which systems are at risk?
- What trends are emerging?
- What will likely happen next?
- What actions should be taken?

It acts like a **virtual Security Operations Center (SOC)** where security analysts can monitor the health of an organization's network from one dashboard.

Domain

Primary Domain

Data Analytics

Supporting Technologies

- Machine Learning
- Artificial Intelligence
- Web Development

- Data Visualization
 - Cybersecurity
-

Problem Statement

Organizations generate millions of network logs every day.

Security teams cannot manually analyze all this information.

Existing tools usually:

- only detect attacks
- produce complex logs
- provide limited explanations
- require experienced analysts

This project solves those problems by converting raw cybersecurity data into meaningful insights through interactive analytics and AI-powered recommendations.

Dataset

Dataset Used

UNSW-NB15

Files

UNSW_NB15_training-set.csv

UNSW_NB15_testing-set.csv

Technologies

Backend

- Python

- Flask

Frontend

- HTML
- CSS
- JavaScript

Database

- MySQL / SQL Server

Machine Learning

- Scikit-Learn
- XGBoost
- Random Forest

AI

- Explainable AI
- AI-generated Security Reports

Visualization

- Plotly
- Chart.js

Development

- VS Code
- Git
- GitHub

Overall Workflow

Dataset

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Data Cleaning

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Data Analytics



Machine Learning



Threat Detection



Explainable AI



Risk Analysis



Threat Prediction



AI Recommendations



Interactive Dashboard

What Does This Project Do?

Imagine a company with

- 500 Employees
- 300 Computers
- Multiple Servers
- Thousands of Network Connections

Every second,

their systems generate security logs.

Instead of manually checking those logs,

this platform automatically analyzes everything.

The system

Collects security data



Analyzes traffic



Finds suspicious activities



Detects attacks



Predicts future attacks



Calculates risk



Provides AI recommendations



Generates reports



Displays everything on a modern dashboard

Main Modules

1 Authentication

Features

- Login
 - Registration
 - Password Encryption
 - User Profiles
-

2 Dashboard

Displays

- Security Score
 - Threat Count
 - Risk Score
 - Active Alerts
 - Recent Activities
 - Today's Attacks
 - Attack Timeline
-

3 Data Analytics

Analyzes

- Dataset Overview
- Attack Distribution
- Protocol Analysis
- Service Analysis
- Traffic Analysis
- User Activity
- Port Analysis
- Network Statistics
- Trend Analysis

Charts

- Pie
 - Line
 - Bar
 - Heatmap
 - Histogram
 - Treemap
 - Network Graph
-

4 Machine Learning

Detects

Normal Traffic

or

Attack

Attack Types

- Generic
- DoS
- Exploit
- Fuzzers
- Reconnaissance
- Worm
- Shellcode
- Backdoor

Models

- Random Forest
- Decision Tree
- XGBoost
- Isolation Forest

Evaluation

- Accuracy
- Precision
- Recall
- F1 Score
- ROC Curve

- Confusion Matrix
-

5 Explainable AI

Instead of saying

Attack

the system explains

WHY

Example

High Packet Rate

Repeated Login Attempts

Abnormal TCP Connections

Suspicious Service Requests

Unknown Protocol Usage

6 AI Attack Story

Automatically generates

A complete incident report.

Example

"At 09:15 AM, the system detected an abnormal increase in HTTP traffic targeting Server A. The machine learning model classified the activity as a Denial-of-Service attack with high confidence. The main indicators were excessive packet rates, repeated requests from the same source IPs, and unusual traffic patterns. The recommended actions are to block the malicious IP addresses, enable rate limiting, and continue monitoring."

7 AI Security Advisor

Provides recommendations such as

- Block malicious IP
 - Enable Multi-Factor Authentication
 - Update firewall rules
 - Change passwords
 - Monitor affected devices
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8 Risk Intelligence

Calculates

Risk Score

for

Users

Devices

Servers

Departments

Overall Organization

Example

Server A

96/100

Critical

9 Threat Investigation

Allows users to investigate

- IP Address
- Device

- User
- Attack Type
- Timeline

Shows complete attack history.

10 Prediction Engine

Predicts

- Future attacks
 - Risk trends
 - Increasing attack categories
 - Security health
-

11 Reports

Generate

- PDF Reports
- CSV Reports
- Excel Reports

AI Executive Summary

Dashboard Snapshot

12 Admin Panel

Manage

- Users
- Dataset
- Models
- Roles
- Logs
- Permissions

Major Features

Analytics

- Dataset Analysis
- Statistical Analysis
- Trend Analysis
- Interactive Dashboard
- Data Visualization

Machine Learning

- Threat Detection
- Attack Classification
- Anomaly Detection
- Model Comparison

Artificial Intelligence

- Explainable AI
- AI Incident Report
- AI Security Advisor
- Executive Summary

Prediction

- Threat Forecast
- Risk Prediction
- Trend Prediction

Risk Management

- Security Score

- Risk Score
 - Device Ranking
 - Department Ranking
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Reporting

- Export Reports
 - Dashboard Snapshots
 - AI Summary
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User Interface

- Glassmorphism Design
 - Light Cyber Theme
 - Responsive Layout
 - Animated Dashboard
 - Interactive Charts
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Expected Outputs

The system provides

- Threat Detection Results
 - Security Dashboard
 - Risk Scores
 - Attack Trends
 - AI Explanations
 - Future Threat Predictions
 - Executive Reports
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Project Architecture

Dataset

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Analytics Engine

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Machine Learning Engine

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AI Engine

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Risk Engine

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Prediction Engine

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Dashboard

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Reports

Estimated Development Timeline

Since **only you** will be developing the project (the team names are only for academic submission), a realistic timeline is approximately **6 months (about 180 days)**.

Month 1

- Requirement analysis
- Research
- Literature survey
- UI/UX design
- Project setup
- Database design

Month 2

- Authentication
- Frontend development
- Dashboard
- Analytics module
- Data preprocessing

Month 3

- Machine Learning model development
- Model evaluation
- Threat detection
- Performance optimization

Month 4

- AI modules
- Explainable AI
- AI Attack Story
- AI Security Advisor
- Risk Intelligence

Month 5

- Prediction engine
- Reports
- Admin panel
- Testing
- Bug fixing

Month 6

- UI polishing
 - Performance improvements
 - Documentation
 - Deployment
 - Presentation preparation
 - Final testing
 - Viva preparation
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Estimated Project Size

Component	Estimate
Python Files	80–100
HTML Templates	15–20
CSS Files	10–15
JavaScript Files	15–20
SQL Scripts	10–15
Total Source Code	~15,000–25,000 lines
Development Time	~6 Months
Primary Developer	4

Final Goal

The final product is **not just an intrusion detection model**. It is a **complete AI-powered Cybersecurity Threat Analytics Platform** that transforms raw network security data into meaningful insights, predicts future threats, explains attack decisions, calculates organizational risk, and helps security teams make informed decisions through an interactive web-based dashboard.

It combines **Data Analytics, Machine Learning, Artificial Intelligence, and modern web technologies** into a single integrated platform suitable as a comprehensive final-year engineering project and a strong portfolio project.